MILITARY SPECIFICATION

OSCILLATOR, HIGH-VOLTAGE: 10549785

1. SCOPE

1.1 This specification covers one type of transistorized, high-voltage oscillator which is a solid-state electronic device for converting 5.90 volt dc (nominal) to 2700 volt ac (nominal) peak-to-peak output at a frequency of 1.2 to 2.0 Kilohertz (KH_z).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-1/1049	6914 Image Converter Tube
MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electrical and Electronic Component Parts

DRAWINGS

U. S. Army, Frankford Arsenal

B10549785	Oscillator, High-Voltage: 10549785

PACKAGING DATA SHEET

10549785	Packaging of Oscillator, High-Voltage

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The oscillator shall be manufactured in accordance with Drawing B10549785.

3.1.1 Moisture-fungus resistance.- The oscillator shall be treated per Type 1 for moisture and fungus resistance in accordance with MIL-T-152.

3.2 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Dimensions and tolerances.
(b) Inorganic protective surface finishes.
(c) Part identification and marking.
(d) Workmanship
(e) Soldering electrical connections.

3.3 Environmental service conditions.

3.3.1 Vibration.- The oscillator shall show no evidence of failure when subjected to vibrations in each of two planes. One plane parallel to the longitudinal axis of the oscillator and the second plane perpendicular to the longitudinal axis of the oscillator. The vibrations shall consist of the following conditions:

(a) An applied frequency of 30 cps, amplitude of .0625 inch (total excursion .1250 inch) for three minutes in each of the two planes.

(b) An applied frequency of 50 cps, amplitude of .04 inch (total excursion of .08 inch) for 2 minutes in each of the two planes.

(c) The applied frequency shall vary uniformly between the limits of 10 and 60 cps at an amplitude of .05 inch (total excursion of .10 inch). The frequency shall be varied from 10 cps to 50 cps and returned to 10 cps in one minute. Repeat this frequency sweep 10 times in each of the two planes. During vibration, the oscillator shall exhibit an output voltage of 2700 ± 100 volts (V) peak-to-peak when operating with 5.90 ± .01 volt dc input, no output load and subjected to a capacitive load of 25 pico picofarads (pf). Subsequent to vibration, the oscillator shall meet the requirements of 3.4.

3.3.2 Shock.- The oscillator shall withstand 6 shock impulses applied in each of 2 mutually perpendicular directions. The acceleration of each shock impulse shall be 100 gravity units (g). The nominal value for pulse duration shall be 3 milliseconds. This requirement shall be met while being subjected to the following conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

(a) Input voltage 5.90 ± .01 volts dc.
(b) No output load and a total capacitive load of 25 pf.
(c) Output voltage 2700 ± 1.00V peak-to-peak.

Subsequent to the shock impulses, the oscillator shall show no evidence of failure and shall meet the requirements of 3.4.

3.3.3 Thermal shock.- While operating with an input of 5.90 ± .01 volt dc, no output load and a total capacitive load of 25 pf, the oscillator shall not be damaged and shall produce an output voltage of 2700 ± 100V peak-to-peak during and at the conclusion of 10 of the following cycles:

(a) Ambient temperature (75°F ± 5°F to plus 150°F).
(b) Plus 150°F to ambient temperature.
(c) Ambient temperature to minus 65°F.
(d) Minus 65°F to ambient temperature.

The rate of change of ambient temperature shall not be less than 108°F per minute.

3.3.4 Operating temperatures.- The oscillator shall be capable of meeting the requirements of 3.4.1, 3.4.2, 3.4.3 and 3.4.5 when thermally stabilized at temperatures ranging from minus 65°F to plus 150°F.

3.3.5 Storage.- The oscillator shall show no evidence of physical or electrical failure subsequent to being stored for 24 hours at temperatures ranging from minus 80°F to plus 160°F.

3.3.6 Humidity, non-operating.- The oscillator shall show no evidence of physical damage or electrical failure due to exposures in atmospheres of 100% RH including condensation.

3.4 Performance.

3.4.1 Operation.- With input voltages of 5.80 ± .01, 5.90 ± .01 and 6.00 ± .01 volts dc, and under all load and source conditions described in Table 1, the oscillator shall operate as follows:

(a) An input current not to exceed 20 milliamperes (ma) dc and a ripple current content of a frequency equal to the operating frequency of the oscillator and whose peak value equals the dc value.

(b) Output voltages as shown in Table 1.

(c) Output frequency of 1.2 to 2.0 kilohertz (KH_z).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

TABLE 1 - Output Voltages

Source Impedances (OHMS)	Output Load d-c	Total Capacitance of Output Load Picofarad (pf)	AC Output Voltage Volts (Peak-to-Peak
0	No Load	25	2700 ± 100
0	No Load	35	2700 ± 100
0	No Load	50	2700 + 100 - 150
0	Full Load	25	2700 ± 100
Max.	No Load	50	1600 ± 200

3.4.2 Rise time.- The oscillator shall start and reach steady state conditions within 5 seconds under the following conditions:

(a) An input voltage of 5.80 ± 0.01 volts dc.
(b) Overload - With the image converter tube energized the amount of radiation on the photocathode is adjusted to 30 milliwatts of power in the image converter.
(c) A total capacitive load of 50 pf.
(d) A source impedance of from zero to maximum.
(e) At a temperature of minus 65°F.

3.4.3 Output current.- With an input voltage of 6.00 ± 0.01 volts, a total capacitive load of 50 pf and no output load, the oscillator's steady state output current shall not exceed 10 milliamperes peak-to-peak and shall contain current surges due to capacitor charging and diode recovery not greater than 200 microamperes total.

3.4.4 Noise level.- While operating at an input voltage of 6.00 ± 0.01 volts, the dc oscillator noise shall be inaudible to a person with normal hearing at a distance of 30 feet against a background noise level of 30 decibels above referenced noise when operated from no load to full load. (Reference level for noise measurement; 100 cps sound wave having intensity of 10^{-16} watt per square centimeter.)

3.4.5 Stray field.- With an input voltage of 6.00 ± 0.01 volts dc, the oscillator shall produce no evidence of stray field (leakage magnetic flux).

3.5 Operational safety.

(a) The oscillator with full output load shall not be damaged when subjected to a reversed polarity input voltage of 7 volts dc for 1 minute.
(b) The oscillator shall start, operate and not be damaged when subjected to the input voltage of 7 volts dc and the output:

(1) Short circuited for 1 minute.
(2) Open circuited for 15 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

(c) The oscillator shall start, operate continuously and not be damaged with an input voltage of 5.90 ± 0.01 volts dc and the output with an overload and total capacitive load of 50 pf. The dc input current shall then not exceed 20 ma and the output peak-to-peak voltage shall be 2500 volts ± 300 volts.

3.6 Reliability.- With an input voltage of 7.0 volts dc the oscillator shall exhibit a minimum mean time between failures (MTBF) of 2000 hours with an acceptable confidence level of 90 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of at least three (3) oscillators (see 4.3.2.2). The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 First article test.- The test specified in 4.6.14 shall apply only to first article testing. Testing to demonstrate the reliability specified in 3.6 (2000 hours MBTF at a 90% confidence level) requires that an oscillator be subjected to a total demonstration testing time of 4600 hours without a failure. To minimize the overall test time duration, the table below is provided to permit equivalent testing concurrently for the total cumulative test time of 4600 hours, without a failure, as follows:

No. Test Items	Cumulative Test Hours	No. Failures Permitted
3	1,533.3 (63.9 days)	0
5	920 (38.3 days)	0
10	460 (19.2 days)	0

NOTE: Selection of additional oscillators to supplement the first article sample for equivalent testing to determine reliability is permitted. Those additional oscillators will be identified and only subjected to the test in 4.6.14.

4.3.2.3 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II, III, IV and V herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II, III, IV and V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: 100% Inspection			
1.	Operation (with 5.9 Vdc input, zero ohm resistance no output load, 50 pf capacitance of output load).	3.4.1	4.6.8
MAJOR: AQL 1.0% Defective			
101.	Vibration	3.3.1	4.6.2
102.	Operation	3.4.1	4.6.8
103.	Output current	3.4.3	4.6.10
104.	Noise level	3.4.4	4.6.11
105.	Stray field	3.4.5	4.6.12
106.	Moisture-fungus resistance	3.1.1	MIL-T-152
MINOR: NONE			

NOTE: The tests for Characteristics 1 through 106 inclusive shall be conducted at a temperature between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One oscillator shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawing - visual
302.	General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental "A".- Three oscillators shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

TABLE IV

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303. Operating temperature (-65°F and +150°F)	3.3.4	4.6.5
304. Rise time (-65°F)	3.4.2	4.6.9

4.4.3 Environmental "B" and operational safety.- In addition to all other sampling specified herein, 3 oscillators shall be randomly selected at the 25, 50 and 75% point of production. These 3 oscillators shall meet the requirements and tests in Table V.

TABLE V

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
305. Shock	3.3.2	4.6.3
306. Thermal shock	3.3.3	4.6.4
307. Storage	3.3.5	4.6.6
308. Humidity, non-operating	3.3.6	4.6.7
309. Operational safety	3.5	4.6.13

4.4.4 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

ITEM		DESCRIPTION
1.	Power supply, dc	1% regulation, 100 mv ripple, to supply input voltages as specified.
2.	Ammeter, dc	Accuracy ± 3% of reading.
3.	Voltmeter	Accuracy ± 0.01% of reading.
4.	Capacitor	Total capacitance of 50 picofarads (includes any capacitance of external components) ± 1%.
5.	Oscilloscope	Accuracy ± 1% of reading (peak-to-peak).
6.	Resistor	0-200 ohms.
7.	Image converter tube	6914 image converter tube conforming to MIL-E-1.

NOTE: Electronic equipment used to energize or measure the performance of the oscillator under test shall not be subjected to the applicable conditions of the test.

4.6 Test methods and procedures.

4.6.1 Moisture - fungus resistance.- The oscillator shall be inspected in accordance with the acceptance inspection procedures of MIL-T-152. The oscillator shall conform to the requirements of 3.1.1.

4.6.2 Vibration.- Standard vibrating equipment capable of providing the amplitudes and frequencies specified in 3.3.1 shall be utilized for the vibration test. The vibration test equipment shall be in conformance with the "Test Facilities" requirements of MIL-F-13926. Place the oscillator in the test housing shown in Figure 1 and attached it securely to the vibration table for vibration in one of the specified planes. Energize the oscillator as specified in 3.3.1. Subject the energized oscillator to the vibrations specified in 3.3.1(a), (b) and (c) for the time period indicated. During these vibrations the output voltage of the oscillator shall conform to the requirements of 3.3.1. Position the oscillator on the vibration table for vibration in the second required plane. Again subject the energized oscillator to the specified vibrations. During these vibrations the output voltage of the oscillator shall again conform to the requirements of 3.3.1. Subsequent to the vibration tests the oscillator shall be removed from the test housing and subjected to a visual and tactile examination for damage. Upon successful completion of the examination for damage

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

the oscillator shall be subjected to the test in 4.6.8, and 4.6.10 through 4.6.12 inclusive to determine conformance to 3.3.1.

4.6.3 Shock.- The oscillator shall be tested in accordance with Method 202 of MIL-STD-202. Place the oscillator in the test housing shown in Figure 1. Attach the test housing to the carriage of the shock testing apparatus so that the longitudinal axis of the oscillator is in a horizontal plane. Energize the oscillator as specified in 3.3.2. Subject the energized oscillator to the six shock impulses specified in 4.4.2. Rotate the test housing on the shock test carriage so that the longitudinal axis of the oscillator is in a vertical plane. Again subject the energized oscillator to the six shock impulses. During these 12 shock impulses (6 shocks each in 2 directions) the output voltage of the oscillator shall conform to the requirements of 3.3.2. Subsequent to the shock tests the oscillator shall be removed from the test housing and subjected to a visual and tactile examination for damage. Upon successful completion of the examination for damage the oscillator shall be subjected to the tests in 4.6.8, and 4.6.10 through 4.6.12 inclusive to determine conformance to 3.3.2.

4.6.4 Thermal shock.- The environmental test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.3. Place the oscillator in the test housing shown in Figure 1. Subsequent to a stabilization period of not less than 30 minutes at 75°F ± 5°F (room temperature) energize the oscillator with voltage specified in 3.3.3. Monitor and record the output voltage of the oscillator during 10 consecutive thermal shocking cycles. A thermal shocking cycle shall consist of the following:

a. Subject the energized oscillator to an ambient temperature of plus 150°F and hold it at this temperature for not less than 30 minutes.

b. At the end of the 30 minute period remove the oscillator from the hot chamber to room temperature and hold it at this temperature for not less than 30 minutes.

c. At the end of the 30 minute period subject the energized oscillator to an ambient temperature of minus 65°F and hold it at this temperature for not less than 30 minutes.

d. At the end of the 30 minute period remove the oscillator from the cold chamber to room temperature and hold it at this temperature for not less than 30 minutes, while continuing to monitor the output voltage. Upon completion of the 10 consecutive thermal shock cycles the oscillator shall be removed from the test housing and subjected to a visual and tactile examination for damage. The oscillator shall conform to the requirements of 3.3.3.

4.6.5 Operating temperatures.- The test equipment specified in 4.6.4 shall be utilized for this test. Place the oscillator in the test housing shown in Figure 1, then place the housed oscillator in the environmental chamber. Connect the oscillator to the test circuit shown in Figure 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

Gradually (not less than 2 hours) reduce the temperature to minus 65°F and hold it at this temperature for not less than 30 minutes. At the end of this 30 minute period energize the oscillator and subject it to the tests in 4.6.8 through 4.6.10, and 4.6.12. Upon successful completion of these tests gradually (not less than 2 hours) raise the temperature to room ambient. Place the housed oscillator in the environmental chamber and connect it to test circuit as specified above. Gradually (not less than 2 hours) raise the temperature to plus 150°F and hold it at this temperature for not less than 30 minutes. At the end of this 30 minute period energize the oscillator and subject it to the tests in 4.6.8 through 4.6.10 and 4.6.12. Upon successful completion of these tests gradually (not less than 2 hours) reduce the temperature to room ambient. Remove the oscillator from the chamber and test housing and visually examine the oscillator for any evidence of damage. The oscillator shall conform to the requirements of 3.3.4 while operating at each temperature extreme.

4.6.6 Storage temperature.- The environmental test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.5. Place the unprotected oscillator (without test housing) in the test chamber and gradually (not less than 2 hours) reduce the temperature to minus 80°F. Hold the oscillator at this temperature (-80°F) for a period of not less than 24 hours. At the end of this 24 hour period gradually (not less than 2 hours) raise the temperature to room ambient. Remove the oscillator from the chamber and examine it for damage. Again place the unprotected oscillator in the test chamber and gradually (not less than 2 hours) raise the temperature to plus 160°F. Hold the oscillator at this temperature (+160°F) for a period of not less than 24 hours. At the end of this 24 hour period gradually (not less than 2 hours) reduce the temperature to room ambient. Remove the oscillator from the chamber and examine it for damage. Subsequent to the visual-tactile examination for damage, place the oscillator in the test housing and connect the oscillator to the test circuit, then subject it ot the test in 4.6.10. The oscillator shall meet the requirements of 3.3.5.

4.6.7 Humidity, non-operating.- Perform this test in accordance with the Humidity Procedure 1 of MIL-F-13926, except that the 12 hour cycle shall be repeated 10 times, and the oscillator shall not be placed in its test housing or energized during the humidity cycles. Upon completion of the humidity cycling remove the oscillator and examine it for damage. Subsequent to the visual-tactile examination for damage, place the oscillator in the test housing and connect the oscillator to the test circuit, then subject it to the test in 4.6.10. The oscillator shall meet the requirements of 3.3.6.

4.6.8 Operation.- Place the oscillator in the test housing shown in Figure 1. Connect the oscillator to the test circuit shown in Figure 2. The test shall then be performed in accordance with the following procedure:

a. Adjust the input voltage to 5.80 ± .01 volts dc, with S_1 closed, R_1 set at zero ohms resistance, and no light incident on the photocathode of the image tube.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

b. Adjust C_1 until the total capacitance output load on the oscillator is 25 pf.

c. Measure and record the input current, output voltage, and output frequency.

d. Readjust C_1 until the total capacitance output load on the oscillator is 35 pf.

e. Measure and record the input current, output voltage, and output frequency.

f. Readjust C_1 until the total capacitance output load on the oscillator is 50 pf.

g. Measure and record the input current, output voltage, and output frequency.

h. Readjust C_1 until the total capacitance output load on the oscillator is 25 pf. Adjust the variable light source incident to the photocathode of the image tube until E2 X 12 = 30 milliwatts.

i. Measure and record the input current, output voltage, and output frequency.

j. Set R_1 to 100 ohms resistance, remove all light incident to the photo-cathode of the image tube, and readjust C_1 until the total capacitance output load on the oscillator is 50 pf.

k. Measure and record the input current, output voltage, and output frequency.
The measured input currents, output voltages and output frequencies obtained in c, e, g, i and k above shall conform to the requirements of 3.4.1. Repeat the above test procedures a through k inclusive, with the input voltage adjusted to 5.90 ± .01 Vdc, and then adjusted to 6.00 ± .01 Vdc. The measured input current, output voltage and output frequency of the oscillator shall conform to the requirements of 3.4.1 during the application of the 3 specified input voltages.

4.6.9 Rise time.- Place the oscillator in the test housing shown in Figure 1. Connect the oscillator to the test circuit shown in Figure 2. Stabilize the oscillator at -65°F. Open S_1 and set R_1 at zero ohms resistance. Adjust the input voltage to 5.80 ± .01 volts dc. Adjust C_1 until the total capacitance output load on the oscillator is 50 pf. Adjust the variable light source incident to the photo-cathode of the image tube until E_2 X 1_2 = 30 milliwatts. Close S_1 and record the time required for the oscillator to start and reach a steady-state condition. Open S_1 and set R_1 at 100 ohms resistance. Again close S_1 and record the time required for the oscillator to start and reach a steady-state condition. Make a minimum of 3 rise time measurements on the oscillator with R_1 set at zero ohms then 100 ohms resistance. The rise time measurements of the oscillator shall not exceed the tolerance specified in 3.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

4.6.10 Output current.- Conduct this test in conjunction with the test in 4.6.8. The measured output current of the oscillator shall conform to the requirements of 3.4.3.

4.6.11 Noise level.- This test shall be performed in an area that conforms to the background noise level requirements of 3.4.4. Perform this test with the oscillator positioned external to the test housing. Connect the oscillator to the test circuit shown in Figure 2. Adjust the input voltage to 6.00 ± .01 volts dc, with S_1 open, R_1 set at zero ohms resistance, and no light incident on the photo-cathode of the image tube. Adjust C_1 until the total capacitance output load on the oscillator is 25 pf. Close S_1 and adjust the variable light source for full load (see 4.6.8h), then no load. Repeat the cycle of no load to full load a minimum of two additional times. During these energizing cycles of the oscillator the distance between the listener and the oscillator shall be 30.0 ± 2.0 feet, without obstruction. The oscillator shall conform to the requirements for noise level as specified in 3.4.4.

NOTE: The person conducting the noise level test will be medically certified as to having normal hearing at the time of the test.

4.6.12 Stray field.- Perform this test with the oscillator positioned external to the test housing. Connect the oscillator to the test circuit shown in Figure 2. Place the output terminal of the oscillator within 1/8 inch, and tangent to the photo-cathode flange of the image converter tube. Adjust the input voltage to 6.00 ± .01 volts dc. The image tube shall be operated at 16 kilovolts (kv) during this test. Adjust the light level incident to the photo-cathode of the image tube, and focus the resolution test pattern for best resolution of the line sets on the screen of the image tube. Turn the oscillator on and off five times and observe the resolution test pattern image projected on the screen of the image tube. Evidence of stray field will show as an increase or decrease in resolution between the "ON" cycles of the oscillator. The oscillator shall conform to the requirements of 3.4.5.

4.6.13 Operational safety.- Place the oscillator in the test housing shown in Figure 1. Connect the oscillator to the test circuit shown in Figure 2. The test shall then be performed in accordance with the following procedure:

a. Adjust the input voltage to 7.00 - .01 volts dc, with S_1 open, R_1 set at zero ohms resistance, and the variable light source adjusted for full load (see 4.6.8h).

b. Adjust C_1 until the total capacitance output load on the oscillator is 25 pf.

c. Reverse the polarity of the input voltage and close S_1 for a period of not less than one minute.At the end of this time period open S_1 and correct the polarity of the input voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

d. Adjust the input voltage to 6.00 ± .01 volts dc, the variable light source to no load, an C_1 until the total capacitance output load is 50 pf.

e. Measure the input current, output voltage, output frequency, and output current. These measured characteristics shall conform to the requirements of 3.4.1 and 3.4.3 for the input voltage and load conditions specified in d above.

f. Adjust the input voltage to 7.00 - .01 volts dc with S_1, R_1 and C_1 as specified in a and b above.

g. Apply a direct short circuit between the output terminal and case ground. Close S_1 for a period of not less than one minute. At the end of this time period open S_1 and remove the short circuit.

h. Open the circuit on the oscillator output. Close S_1 and operate the oscillator for a period of not less than 15 minutes. At the end of this time period open S_1 and close the open circuit.

i. Repeat the procedures in d and e above. The oscillator shall again conform to the requirements of 3.4.1 and 3.4.3.

j. Adjust the input voltage to 5.90 ± .01 volts dc, with S_1 open, R_1 set at zero ohms resistance, and the variable light source adjusted for full load (see 4.6.8h). Adjust C_1 until the total capacitance output load on the oscillator is 50 pf.

k. Close S_1 and operate the oscillator for a period of not less than one hour. During this time period moniotor the input current and output voltage. The dc input current and output peak-to-peak voltage shall conform to the requirements of 3.5(c) during this one hour time period. Open S_1.

l. Repeat the procedures in d and e above. The oscillator shall again conform to the requirements of 3.4.1 and 3.4.3.

4.6.14 Reliability.- Connect the oscillator to a test circuit as shown in Figure 2. Adjust the input voltage to 7.0 ± .01 volts dc; R_1 set at zero ohms resistance; the variable light source adjusted to no load; and C_1 adjusted for a total capacitance output load of 50 pf on the oscillator. Operate the oscillator(s) continuously, 24 hours per day, for the total cumulative test hours as specified in 4.3.2.2. At the completion of the total cumulative test hours reduce the input voltage to 6.0 ± .01 volts dc, while maintaining R_1, C_1, and variable light source as specified above, and measure the input current, output voltage, output frequency, and output current. To conform to the requirements of 3.6 the oscillator(s) shall function throughout the total cumulative test hours without a failure, and shall then conform to the applicable input current, output voltage, output frequency and output current requirements specified in 3.4.1 and 3.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking.

Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10549785.

6. NOTES

6.1 Intended use.

The oscillator is intended to provide a high voltage a-c input to the multiplier section of the image intensifier night vision system of the Tank Periscopes XM44E1, XM50 and XM51.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable stock number.
d. Requirement for First Article Testing.

Custodian

Army - MU

Preparing activity

Army - MU

Project No. 1240-A126

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

OSCILLATOR TEST HOUSING

DIMENSIONS APPROXIMATE ONLY
OSCILLATOR TO FIT SNUGLY IN CAVITY

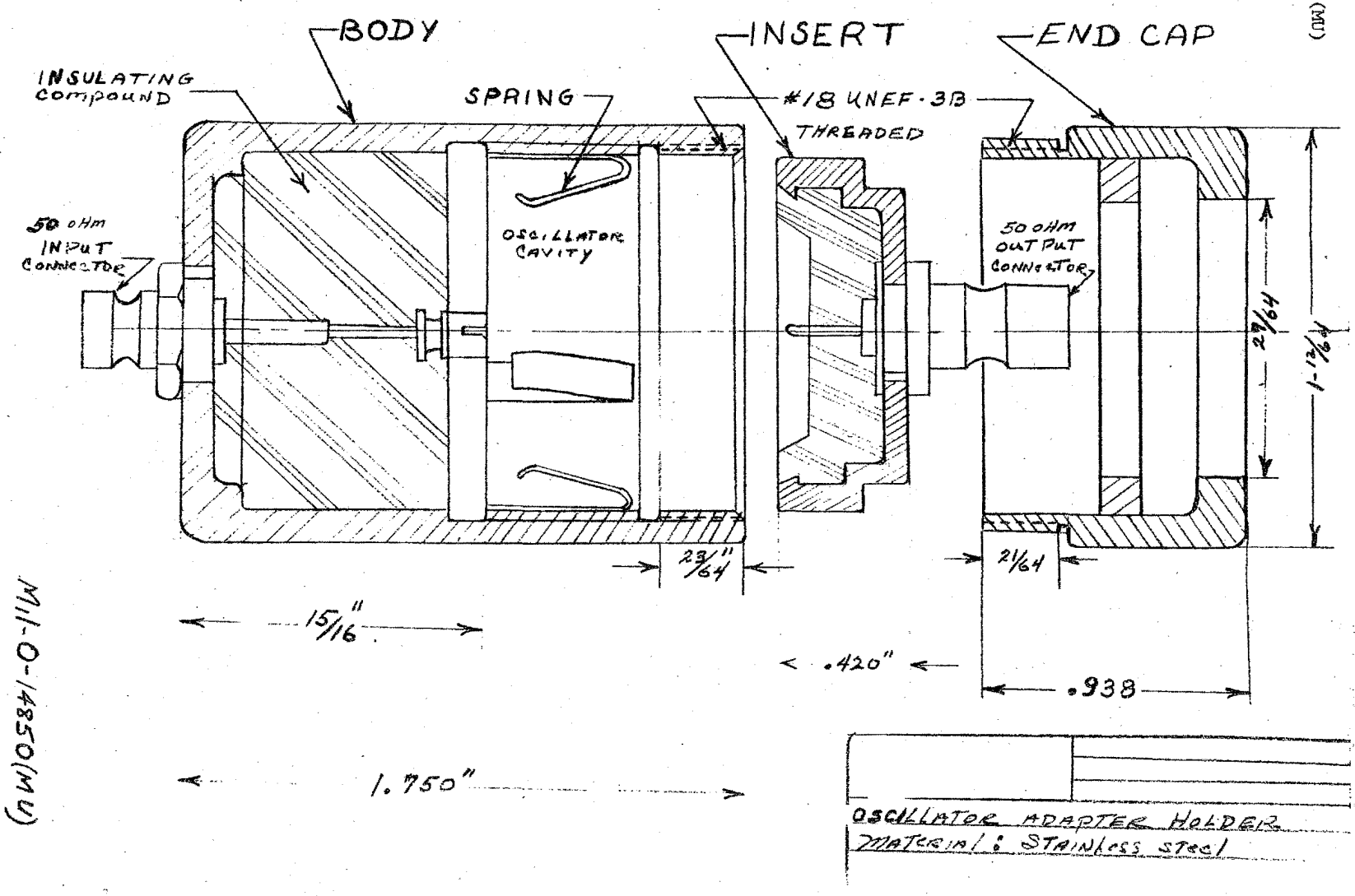


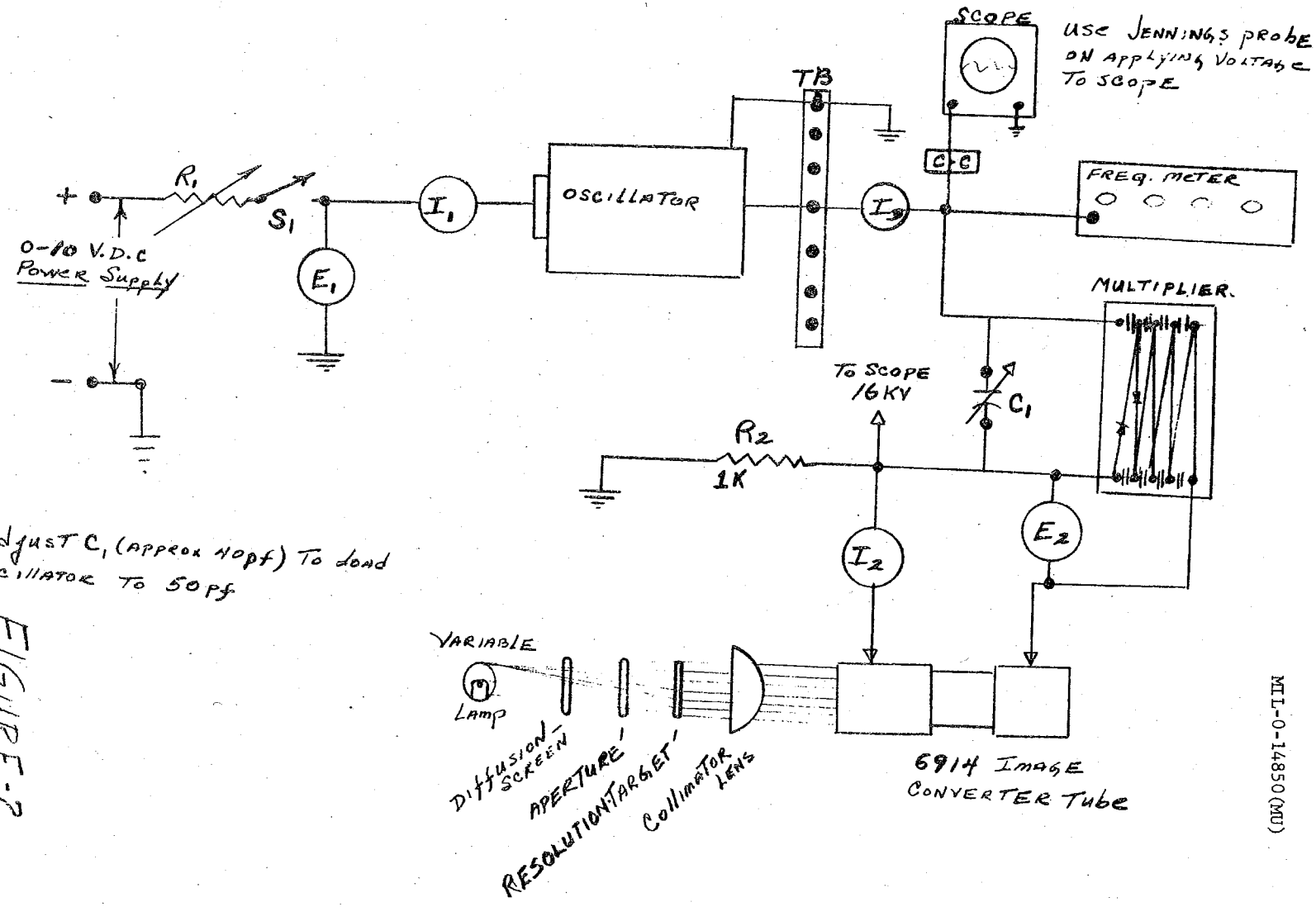
FIGURE-1

MIL-O-14850(MU)

MIL-O-14850(MU)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

Oscillator Test Circuit



1) Adjust C_1 (approx 40pf) To load oscillator To 50pf

FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army

Frankford Arsenal
Philadelphia, Pa. 19137

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
Department of the Army

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-N1100
Philadelphia, Pa. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form *will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments* and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:55Z
Check the source to verify that this is the current version before use.